



A Glass-Box Evidence-Synthesis System for Classical TCM

via Hybrid RAG and Knowledge Graph Integration

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The Problem

Why do TCM practitioners need a new kind of AI tool?

TCM: A Knowledge-Intensive Domain

Traditional Chinese Medicine is built on **thousands of years** of classical literature.

-  Practitioners must master texts spanning from 黃帝內經 (c. 200 BCE) to 溫病條辨 (1798 CE)
-  This requires **years of study** — book knowledge + clinical experience
-  AI can learn from these texts and retrieve the precise details **in seconds**

 **Core Idea:** What if AI could serve as an intelligent assistant that retrieves exactly the right classical passage when a practitioner needs it?

② Why Not Just Ask Existing AI?

Several TCM-specialized LLMs already exist — Qihuang (岐黄), HuatuoGPT, TCMChat...

None provide passage-level citation tracing to classical sources.

- ⚠️ You get an answer, but **where did it come from?**
- ✖️ The practitioner **cannot verify** against the original text
- 🗝️ It's a **black box** — plausible-sounding, but unverifiable

👁️ TCM-Sage is a **glass box**: every claim is traceable to a specific passage in a specific classical text, with knowledge graph context linking related concepts.

Why General AI Falls Short

The limitations of web search for domain knowledge

The Trust Problem with AI + Web Search

1. AI投毒 (AI Poisoning)

Fake content published online gets recommended by AI within **minutes**

→ 茗感神经 experiment (2025): fabricated milk tea brand → AI recommended it as #1

2. Citation Mismatch

AI cites articles where only the **title** is relevant — content is unrelated

3. Phantom Tool Calls

LLM **claims** it searched online but never actually invoked the search tool

4. Graceful Fabrication

When AI can't find something, it invents **plausible-sounding excuses** instead of admitting "I don't know"

These are not hypothetical risks — each has documented real-world evidence (see Appendix: AI投毒 Evidence).

TCM-Sage's Approach

❌ General LLM + Web Search

- ⚠️ Searches unreliable internet sources
- 🦠 Vulnerable to AI Poisoning
- 💠 Cannot guarantee source authenticity
- ↓ Answer quality depends on what's online

✅ TCM-Sage (RAG)

- 📖 Retrieves from **curated classical texts** verified over millennia
- 🔗 Context injected **directly** into the prompt
- 🔍 LLM **synthesizes** — never searches the web
- 🔍🟢 Every claim traceable + **KG** linking related concepts

📘 When RAG provides context, the LLM does not hallucinate. The failure mode is RAG not finding what the user needs — not the AI making things up.

This architectural choice directly addresses the trust problem from the previous slide.



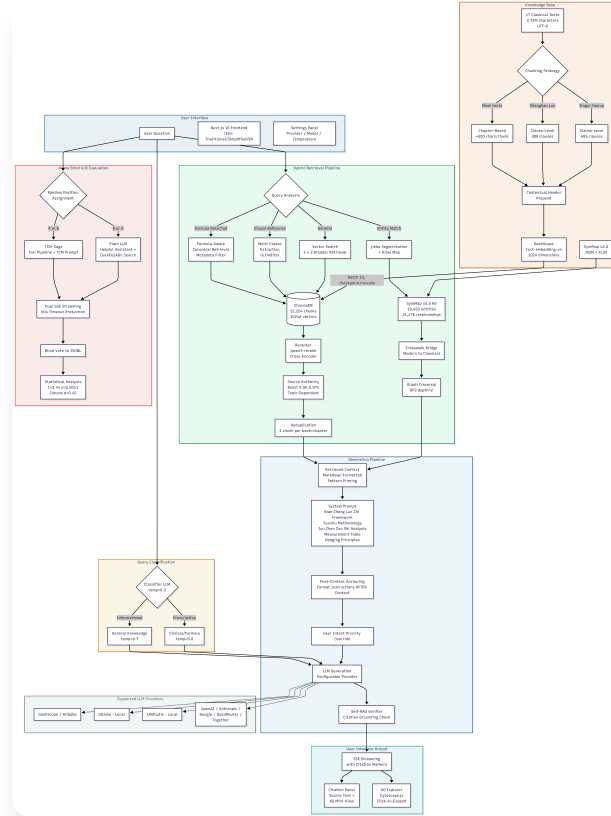
Building TCM-Sage

From a single text to a production system

How I Got Here

-  Originally selected **AI-powered course selection** as my FYP topic
-  My dad suggested pivoting to **Traditional Chinese Medicine** — he had connections to practitioners
-  I also had **friends studying TCM** who confirmed my interest
-  Asked Dr. Zhang Ce to change topic — **approved**
-  The thread throughout: **AI** — just the application domain changed

System Architecture



Query → Classification → Hybrid Retrieval (Vector + KG) → Reranking → Generation → Verification

Phase 1: Foundation

September 2025 — January 2026

What I Built

-  **1 text:** 黃帝內經 (Huangdi Neijing)
-  CLI prototype → **Streamlit** web UI
-  Hand-built KG entities (for stability)
-  3-layer AI extraction pipeline — ready but unverified

Mid-Point Demo (Jan 2026)

- Demoed Streamlit prototype
- CLI mentioned as early-stage prototype

Prof. Wang's Challenge:

"The student is encouraged to think about the key difference between it and existing LLM models."

Phase 2: Responding to the Challenge

February — March 2026



Knowledge Graph Pivot

3-layer AI pipeline was ready → Dr. Zhang advised **credible academic source** → Pivoted to **SymMap 2.0** (18,450 entities, 21,476 relationships) + **crosswalk bridge** for ancient ↔ modern terms



UI Pivot

Streamlit → **Next.js 16 production frontend** — inspired by Kenny's vision of building a **platform**



The Breakthrough: 17 Foundational Texts

Kenny (HKBU SCM Year 5, cGPA 3.97) helped select the texts. 1 → **17 classical TCM texts** (3.72M characters). Even concepts from 黃帝內經 were explained **better** with the full theoretical foundation.

Domain-Specific Retrieval



Clause-Level Chunking

Regular embeddings couldn't handle "傷寒論第82條" searches. **388 + 489** clauses individually indexed.



Embedding Upgrade

all-MiniLM (384d) → nomic (768d) → **DashScope text-embedding-v4** (1024d) with TCM domain-specific prefixes.



Reranking

qwen3-rerank in hybrid search pipeline. Most relevant passages first.

Technical Contributions

These five points summarize the project's core technical contributions.

1 Hybrid RAG Pipeline for Classical Chinese

Vector search + SymMap 2.0 KG + crosswalk bridge for ancient ↔ modern terms

2 Domain-Specific Retrieval Optimizations

Clause-level chunking, formula-aware canonical retrieval, source authority boosting

3 RAG Context Engineering

Pattern Priming (markdown context → markdown output), post-context instruction anchoring

4 Arena Blind A/B Evaluation

Dual-streaming evaluation with paired t-test and Cohen's d validation

5 Practitioner Validation

Real-world testing by TCM students and practitioners from 3 institutions

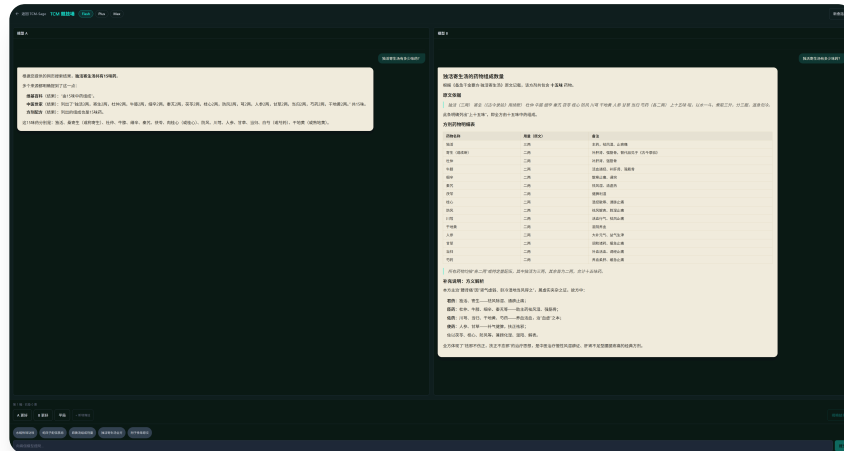


Proving It Works

Arena Blind Evaluation & Statistical Results

Arena: Blind A/B Evaluation

- 💡 Idea born in a **catch-up meeting** with Dr. Zhang
- 📈 Grew into full implementation inspired by **LMSYS Chatbot Arena**
- 🗂️ Two responses streamed **side-by-side**, identity hidden
- ✅ **Model A**: TCM-Sage (RAG-enhanced)
- ❌ **Model B**: General AI (LLM + web search)
- 🧑 User votes **without knowing** which is which



Statistical Results

👤 59 Blind Votes (38 RAG / 6 Tie)

From real TCM practitioners and students:

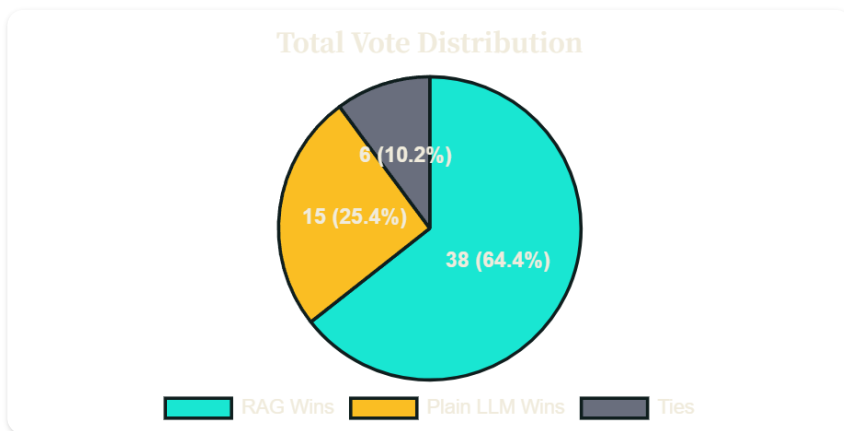
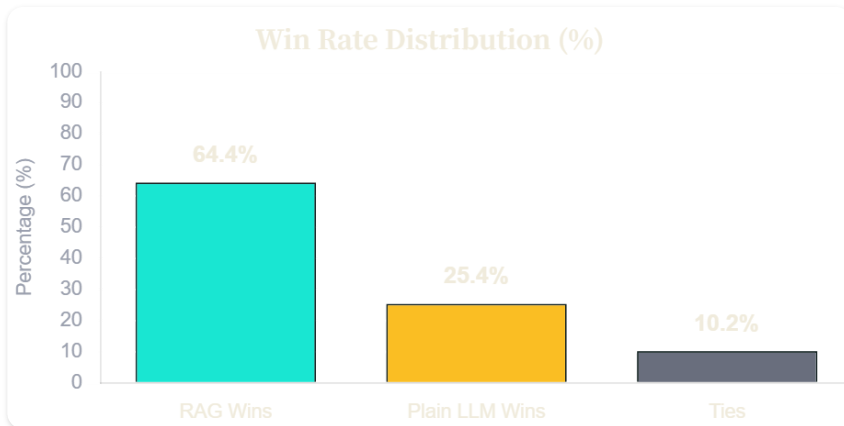
- 🏫 HKBU School of Chinese Medicine
- 🏫 HKU TCM students
- 🎓 Doctoral students at 廣東省中醫藥大學
- 🏥 Practitioners at 廣東省中醫院

Metric	Value
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Paired t-test	$p = 0.0011$
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Cohen's d	0.45 (medium)
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Conclusion	Significant preference for RAG
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☆ Practitioner Feedback

Qualitative feedback from **Kenny Woo Shi Nam** (HKBU SCM Year 5, cGPA 3.97/4.0):



Focused & Relevant

Focused answers with adequate elaboration and valid arguments — no irrelevant details



Credible Sources

Extracts and integrates credible sources from TCM classics



TCM Thinking

Guides practitioners how to think from TCM perspectives — treatment approaches, medica recommendations



Domain-Tuned

Tuned for the TCM domain

→ Kenny has expressed interest in recommending the system to **HKBU School of Chinese Medicine**

 Live Demo

Live Demonstration

1 Main Chat — TCM question → streaming + citation panel

2 Clause Retrieval — "傷寒論第82條" → exact match

3 KG Explorer — Entity relationships in SymMap 2.0

4 Arena Stats — T-test results & charts

5 Settings — Multi-provider LLM switching

 *If live connectivity is unstable, I will continue with prepared screenshots and video of the same workflow.*

Discussion

! Limitations

All three limitations are acknowledged with current status and a clear next-step plan.

🚫 RAG Absence Hallucination

When retrieval fails, LLM fabricates excuses (e.g., "此文献可能已在历史中遗失"). Prompt fix attempted → over-corrected → removed. Original trigger resolved by clause-level chunking; underlying issue **open but non-reproducible**.

📖 Corpus Scope

17 texts cover foundational theory well. Modern clinical practice material still **limited** (practitioner feedback).

📊 Sample Size

59 votes — small but statistically significant ($p < 0.01$). Quality of testers (real practitioners) matters alongside quantity.

Future Work



Corpus Expansion

Practitioner-recommended: 张锡纯《医学衷中参西录》, works by 刘渡舟, 冯世伦, modern acupuncture texts



Patient-Friendly Mode

Simplified explanations alongside classical citations for non-specialist users



Platform Development

Kenny interested in HKBU SCM recommendation. Each institution maintains own knowledge base.



Fine-Tuned Base Model

Stack TCM-specialized LLM (e.g., HuatuoGPT) + RAG pipeline for compounding improvements

✓ Conclusion

- ✓ Built a **working evidence-synthesis system** for classical Traditional Chinese Medicine
- ✓ Fills a gap neither general LLMs nor existing TCM AI products fill: **transparent, citation-backed retrieval**
- ✓ **17 classical texts, 12,200+ chunks**, clause-level precision for 傷寒論 and 金匱要略
- ✓ **SymMap 2.0** knowledge graph with crosswalk bridge for ancient ↔ modern term mapping
- ✓ Statistically validated: **p = 0.0011**, tested by **real TCM practitioners**
- ✓ **Glass box, not black box** — every answer is verifiable

Glass-box verifiability — where every answer traces back to a classical source — is what sets TCM-Sage apart.

Thank You 

Questions?

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Appendix

Why RAG, Not Fine-Tuning?

	🔍 RAG	🤖 Fine-Tuning
Transparency	Can see retrieved sources	Black box
Training data	None needed	Requires instruction-tuning pairs
Corpus updates	Add text files, re-ingest	Retrain the model
Base LLM	Swappable (8 providers)	Locked to one model
Classical Chinese	Works with any LLM	May degrade general capabilities

📘 RAG provides the transparency that TCM practitioners need to **verify** AI-generated advice against original texts.

Sample Size: Is 59 Enough?

- ✓ **p = 0.0011** — well below 0.05 threshold, even below 0.01
- ✓ **Cohen's d = 0.45** — medium effect size, meaningful practical significance
- ✓ Testers were **real TCM practitioners and students**, not random participants
- ✓ Consistent with LMSYS Chatbot Arena vote-based methodology

Multi-Provider Architecture

8 LLM Providers Supported:

DashScope (Qwen)

OpenAI

Anthropic

Google

OpenRouter

Together AI

Ollama

LMStudio

⚙️ Switch providers via Settings UI — no code changes

🔒 **Local deployment:** Ollama/LMStudio for complete privacy

Query Classification & Dual Temperature



Informational

"什么是六经辨证？"

→ Higher temperature for richer explanation



Prescriptive

"麻黄汤的剂量？"

→ Lower temperature for clinical precision

Classification runs **first** (separate smaller model) → determines temperature → then retrieval + generation

RAG Hallucination: The Full Story

- 1 Kenny searched "傷寒論第82條" → system returned **Clause 2** instead (wrong)
- 2 Implemented clause-level chunking → found correct clause, but **only one at a time**
- 3 "第2條和第82條" → found Clause 2, **missed** Clause 82
- 4 "第82條和第2條" → found 82 correctly (query-ordering bias revealed)
- 5 Implemented **multi-clause search** → fixed
- 6 Deeper issue: when retrieval misses, LLM fabricates "此文献可能已在历史中遗失"
- 7 Prompt fix → **over-corrected** → removed. Issue **open but non-reproducible**

AI投毒 Evidence

茗感神经 Experiment

AI新榜, Sep 2025

Fabricated milk tea brand, 2 fake articles.
Within **10 minutes**, AI recommended it as #1.

Salesforce Research

Mar 2026

"Poisoning the Well" — **~80% attack success rate** on AI search agents.

虎嗅 Investigation

Mar 2026

"当8家顶级AI集体说谎" — fake product across **8 AI platforms**, all fell for it.

✓ TCM-Sage's curated classical corpus is **immune** to this attack vector.

Crosswalk Bridge

❗ Problem

- Same word can mean **different things** across 2000 years
- Modern "感冒" ≠ classical "伤寒"
- Both relate to cold/flu but have different clinical meanings

✓ Solution

- **SymMap 2.0** provides modern terminology (18,450 entities)
- **Crosswalk bridge** maps modern → classical entities
- jieba segmentation + colloquial alias mapping
- A 2025 paper confirmed this is "**largely unexplored**"